

Lesson 10: Transformations

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2025-10-27

Learning Objectives

1. Find the pdf of a linear rescaling of a random variable
2. Find the pdf of a nonlinear transformation of a random variable using the CDF method

Distributions of transformations of random variables

- Often make transformations of RVs
- A function of a random variable is a random variable
 - If X is a random variable and g is a function then $Y = g(X)$ is a random variable
 - Since $g(X)$ is a random variable it has a distribution
- Distribution of $g(X)$ will have a different shape than the distribution of X
- Two types:
 - Linear rescalings: $g(u) = a + bu$
 - Nonlinear transformations: e.g. $g(u) = u^2, g(u) = \log(u)$

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Linear rescaling

Definition: Linear Rescaling

A **linear rescaling** is a transformation of the form $g(u) = a + bu$, where a and b are constants

- Thus, if we have a random variable, X , then a linear rescaling of X could be $M = g(X) = a + bX$
- For example, converting temperature from Celsius to Fahrenheit using $g(u) = 32 + 1.8u$ is a linear rescaling.

Example of linear rescaling (1/4)

Example 1: Linear rescaling of U

Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = 1 - U$

1. What are the possible values of V ?
2. Is V the same random variable as U ?
3. Find $P(U \leq 0.1)$ and $P(V \leq 0.1)$.
4. Find the pdf of V .
5. Does V have the same distribution as U ?

Example of linear rescaling (2/4)

Example 1: Linear rescaling of U

Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = 1 - U$

1. What are the possible values of V ?
2. Is V the same random variable as U ?

1. Possible values of V .

Since $V = 1 - U$ and $U \in [1, 2]$,

$$V \in [1 - 2, 1 - 1] = [-1, 0].$$

2. Is V the same random variable as U ?

No. They are different random variables: U takes values in $[1, 2]$ while V takes values in $[-1, 0]$. (They are deterministically related, but not the same.)

Example of linear rescaling (3/4)

Example 1: Linear rescaling of U

Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = 1 - U$

3. Find $P(U \leq 0.1)$ and $P(V \leq 0.1)$

3. $P(V \leq -0.5)$.

$V \leq -0.5 \iff 1 - U \leq -0.5 \iff U \geq 1.5$. So

$$P(V \leq -0.5) = P(U \geq 1.5) = \int_{1.5}^2 \frac{4}{15} u^3 du.$$

Compute the integral:

$$\int_{1.5}^2 \frac{4}{15} u^3 du = \frac{4}{15} \left[\frac{u^4}{4} \right]_{1.5}^2 = \frac{1}{15} (2^4 - 1.5^4) = \frac{1}{15} (16 - 5.0625) = \frac{10.9375}{15} = \frac{35}{48} \approx 0.72917.$$

Example of linear rescaling (4/4)

Example 1: Linear rescaling of U

Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = 1 - U$

4. Find the pdf of V .

5. Does V have the same distribution as U ?

4. PDF of V .

Use the one-to-one transformation $u = 1 - v$. $|du/dv| = 1$. For $v \in [-1, 0]$,

$$f_V(v) = f_U(1 - v) = \frac{4}{15}(1 - v)^3, \quad -1 \leq v \leq 0,$$

and $f_V(v) = 0$ otherwise.

5. Does V have the same distribution as U ?

No. Their supports differ (U on $[1, 2]$, V on $[-1, 0]$) and their PDFs are different, so they do not have the same distribution.

Standard Normal Example

In general, if Z has a $\text{Normal}(0, 1)$ distribution then $X = \mu + \sigma Z$ has a $\text{Normal}(\mu, \sigma)$ distribution.

1. Is Y the same random variable as X ?
2. Does Y have the same distribution as X ?
3. Donny Don't says that the distribution of $-Z$ will look like an “upside-down bell”. Is Donny correct? If not, explain why not and describe the distribution of $-Z$.
4. Donny Don't says that the standard deviation of $-Z$ is -1. Is Donny correct? If not, explain why not and determine the standard deviation of $-Z$.

Summary

- A linear rescaling of a random variable does not change the basic shape of its distribution, just the range of possible values.
 - It can flip it, widen it, condense it, and/or shift it
- Remember, do NOT confuse a random variable with its distribution
 - The random variable is the numerical quantity being measured
 - The distribution is the long run pattern of variation of many observed values of the random variable

Learning Objectives

1. Find the pdf of a linear rescaling of a random variable

2. Find the pdf of a nonlinear transformation of a random variable using the CDF method

Nonlinear transformations

- What happens when we make a **nonlinear transformation**, like a logarithmic or square root transformation?
- Nonlinear transformations do *not* necessarily preserve the distribution shape
- Examples of nonlinear transformations:
 - $g(u) = u^2$
 - $g(u) = \sqrt{u}$
 - $g(u) = \log(u)$
 - $g(u) = e^u$
 - $g(u) = \frac{1}{u}$

Finding the pdf of a transformation

- Let M be a transformation of X : $M = g(X)$
- When we have a transformation of X , M , we need to follow the **CDF method** to find the pdf of M

We follow **CDF method**:

1. Start with the pdf for X
 - aka $f_X(x)$
2. Translate the domain of X to M : find the possible values of M
3. Find the CDF of M
 - aka $F_M(m) = P(M \leq m) = P(g(X) \leq m)$
 - Will require manipulating $g(X) \leq m$ in terms of X (aka X alone on the left side)
4. Take the derivative of the CDF of M with respect to m to find the pdf of M
 - aka $f_M(m) = \frac{d}{dm} F_M(m)$

Example of nonlinear transformation (1/4)

Example 2: Nonlinear transformation of U

Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = \log(U)$

1. What are the possible values of V ?
2. Find the CDF of V
3. Find the pdf of V

Example of nonlinear transformation (2/4)

Example 2: Nonlinear transformation of U

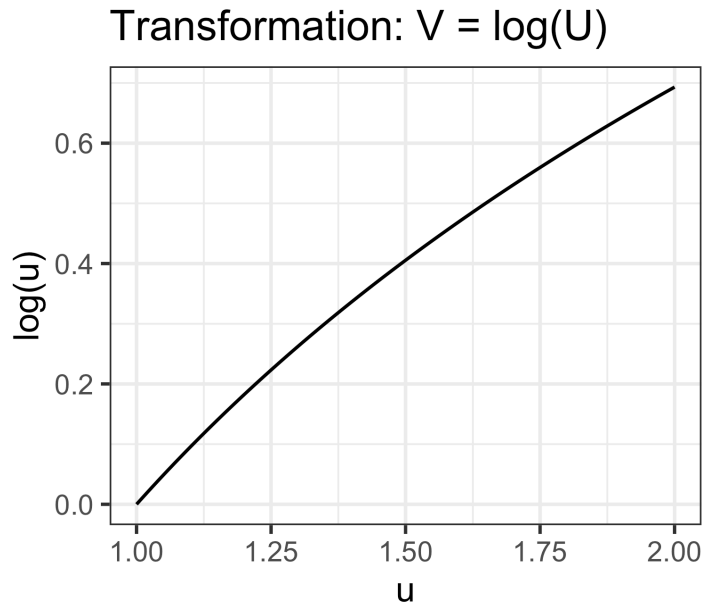
Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = \log(U)$

1. What are the possible values of V ?

1. Possible values of V

Since $U \in [1, 2]$ and $V = \log(U)$,

$$V \in [\log 1, \log 2] = [0, \log 2].$$



Example of nonlinear transformation (3/4)

Example 2: Nonlinear transformation of U

Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = \log(U)$

2. Find the CDF of V

2. CDF of V

For any real v ,

$$F_V(v) = P(V \leq v) = P(\log U \leq v) = P(U \leq e^v).$$

Using the CDF of U , for $1 \leq u \leq 2$,

$$F_U(u) = \int_1^u \frac{4}{15} t^3 dt = \frac{u^4 - 1}{15}.$$

Thus

$$F_V(v) = \begin{cases} 0, & v < 0, \\ \frac{e^{4v} - 1}{15}, & 0 \leq v \leq \log 2, \\ 1, & v > \log 2. \end{cases}$$

Example of nonlinear transformation (4/4)

Example 2: Nonlinear transformation of U

Let U be a random variable with $f_U(u) = 4u^3$ for $1 \leq u \leq 2$. Define $V = \log(U)$

3. Find the pdf of V

3. PDF of V

Differentiate the CDF on $[0, \log 2]$ or use the transformation formula:

$$f_V(v) = f_U(e^v) \cdot \frac{d}{dv} e^v = \frac{4}{15} (e^v)^3 \cdot e^v = \frac{4}{15} e^{4v}, \quad 0 \leq v \leq \log 2,$$

and $f_V(v) = 0$ otherwise.

(You can check normalization: $\int_0^{\log 2} \frac{4}{15} e^{4v} dv = (1/15)(e^{4 \log 2} - 1) = (1/15)(16 - 1) = 1$.)

Example of nonlinear transformation: domain (1/4)

Example 3: Nonlinear transformation of X

Let X be a random variable with $f_X(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$. Define $Y = X^2$

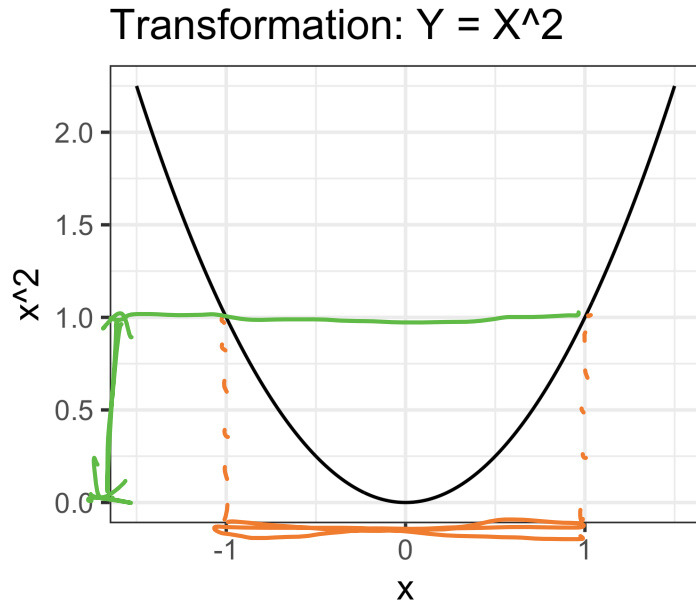
1. What are the possible values of Y ?
2. Find the CDF of Y
3. Find the pdf of Y

Example of nonlinear transformation: domain (2/4)

Example 3: Nonlinear transformation of X

Let X be a random variable with $f_X(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$. Define $Y = X^2$

1. What are the possible values of Y ?



1 Possible values of Y

Since $X \in [-1, 1]$ and $Y = X^2$,

$$Y \in [0, 1].$$

Example of nonlinear transformation: domain (3/4)

Example 3: Nonlinear transformation of X

Let X be a random variable with $f_X(x) = \frac{1}{2}$

2. Find the CDF of Y

2 Find the CDF of Y

For any y ,

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y).$$

For $0 \leq y \leq 1$,

$$P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}).$$

Because X has pdf $f_X(x) = \frac{1}{2}$ on $[-1, 1]$,

$$F_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \frac{1}{2}(2\sqrt{y}) = \sqrt{y}.$$

Thus,

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

Example of nonlinear transformation: domain (4/4)

Example 3: Nonlinear transformation of X

Let X be a random variable with $f_X(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$. Define $Y = X^2$

3. Find the pdf of Y

3 Find the pdf of Y

Differentiate the CDF for $0 < y < 1$:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{2\sqrt{y}}.$$

So

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$